2026_TCS_Digital_Mock Test_3

Test Summary

- No. of Sections: 3
- No. of Questions: 32
- Total Duration: 100 min

Section 1 - Advanced Quantitative Ability

Section Summary

- No. of Questions: 20
- Duration: 40 min

Additional Instructions:

None

Q1. In an exam, Vinesh scored 25 marks less than Suresh. Suresh scored 45 more marks than Vikram. Hari scored 75 marks which is 10 more than Vikram. If Ganesh's score is 50 less than the maximum marks of the test. What approximate percentage of marks did Ganesh score in the exam, if he gets 34 marks more than Vinesh?

- 90
- 70
- 80
- 85

Q2. 2 numbers are 26% and 5% more than the third. What percentage is the first of the second?

- 120%
- 100%
- 80%
- 125%

Q3. Diwana buys a book from her friend Tarun, who sells the book by giving a discount of 20% from the price he fixed. But Tarun, being a cunning mind, increased the price of the book by 50% from the rate at which he bought, only from this price he gave a discount of 20 %. Diwana again sells the same book to her other friend for an extra 20Rs. Now finally this SP is actually 30% more than the original Cost price of the book. So, what is the profit of Diwana? (in percentage)

- 8.33%
- 6%
- 9%

10%

Q4. A shopkeeper bought an item at Rs. 3600 and while selling it offered a 40% discount on the marked price and earned a profit of Rs.1200. If a discount is reduced by 10%, then what is the percentage profit on the cost price?

34.44%

42.22%

48.88%

38.88%

Q5. The frustrum of a right circular cone has the radii of base 5 cm, of the top 3 cm and a height of 6 cm. Find the volume of the frustrum.

204 cm^3

308 cm^3

245 cm^3

250 cm^3

Q6. Two mutually perpendicular chords AB and CD meet at a point P inside the circle such that AP = 6 cms, PB = 4 cms and DP = 3 cms. What is the area of the circle?

100π/7 sq cms

125π/8 sq cms

52π/3 sq cms

125π/4 sq cms

Q7. A bank offers two different schemes at the same rate of interest for investment. A woman chose scheme II to invest a sum of money but due to a fall in the interest rate from 13% to 12 1/2%. her yearly income diminished by ₹104. Calculate the compound interest on the same amount for 2 years at the same rate if she invested in the scheme I.

5757

5762

5759.52

5552.75

Q8. The difference between the simple interest on a certain sum at the rate of 20% per annum for 18 months and compound interest which is compounded every 6 months is Rs.124. What is the principal sum?

Rs.6000

Rs.5000

Rs.8000

Rs.4000

Q9. Clark got Rs.3000 as his share out of a total profit of Rs.4500 which he and Chintu earned at the end of one year. If Clark invested Rs.10000 for 6 months, whereas Chintu invested his amount for the whole year, what was the amount invested by Chintu?

Rs. 1000

Rs. 5000

Rs. 2200

Rs. 2500

Q10. A got married 5 years ago. A's present age is 5/4 times his age at the time of marriage. 4 years from now, A' s wife will be 3/2 times her age at the time of marriage. Difference in age between husband and wife is

2

4

6

8

Q11. P, Q, and R get Rs.1105 for doing a job together. While working alone, P, Q, and R need 7 days, 8 days, and 11 days respectively to complete the job. Then what would be the share of each?

- P = Rs. 440; Q = Rs. 385; R = Rs. 280
- P = Rs. 500; Q = Rs. 400; R = Rs. 205
- P = Rs. 445; Q = Rs. 375; R = Rs. 285
- P = Rs. 446; Q = Rs. 381; R = Rs. 278

Q12. A completes a work in 24 days. B completes it in 30 days. If they are working together and after 10 days B left and remaining work is completed by A alone. Find the wages of B in Rs.7500

- Rs.2500
- Rs.5000
- Rs.4500
- None of these

Q13. Two trains each 120 m long, pass each other on parallel lines. If they are going in the same direction, the faster one takes 16 seconds to pass the other completely. If they are going in different directions, they completely pass each other in 48/13 Seconds. Find the speed of faster train in kmph.

- 125 kmph
- 98 kmph
- 144 kmph
- 160 kmph

Q14. The probability of hitting 120m high six by three different batsmen is 1/3, 1/4, and 1/5 respectively. Find the probability that only one of the batsman will hit 120m six when they hit simultaneously?

- 11/24
- 9/24
- 1/13
- 13/30

Q15.

Out of 16 destinations for a vacation, 7 places are in Asia and 9 in Europe. Two places are to be selected for the vacation. The probability that at least one of the selected places will be in Asia is:

- 7/10
- 4/5
- 6/11
- 2/3

Q16. The Australian government had decided to change the number plate of vehicles to "two letters followed by two digits with the first digit, not zero" instead of the old pattern which had two letters followed by three digits. How many different number plates can be assigned? (Assuming repetition is allowed).

- 75430
- 60840
- 546740
- 87560

Q17. How many different ways are there to select 4 students for a team from 6 students of age 12 and 4 students of age 13, such that at least one student from age 12 group should be present in it?

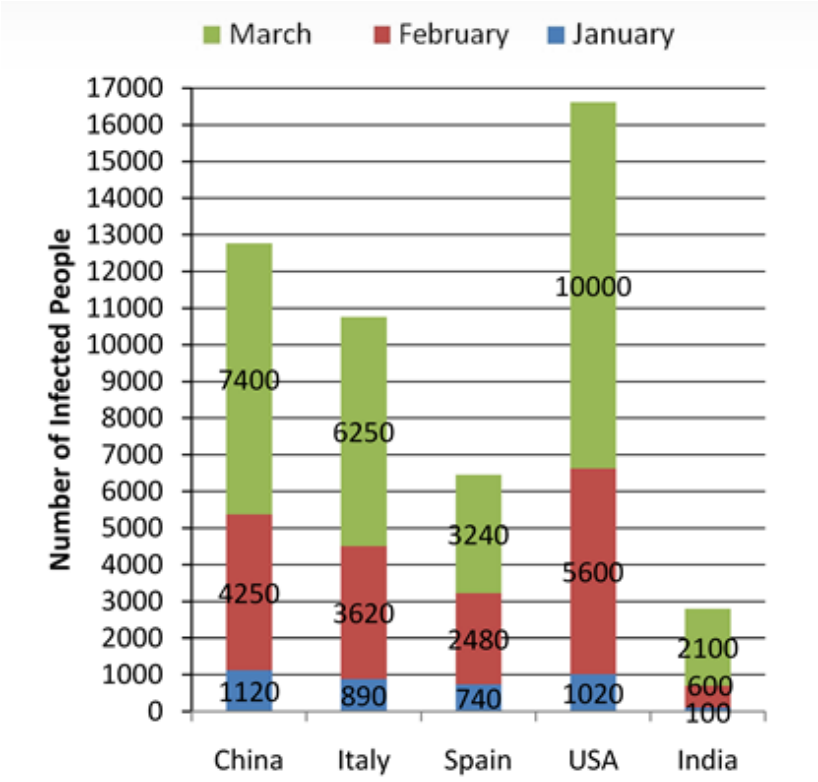
- 209
- 205
- 194
- 159

Common Content:

Study the following bar chart carefully and answer the questions given beside.

Data regarding the number of people who were tested positive with COVID-19 during January, February, and March in five countries is given in the stack bar chart.

Data of number of People in five countries who were found positive for COVID-19 test



Q18. Find the average number of people per day who were tested positive in India in March.

- 60.73
- 67.74
- 72.34
- 76.47

Q19. Japan in January had twice the number of cases that India had in January while 50% more cases in February than India that India had in February. Find the number of cases in Japan in March if cases in March were twice the total cases till February end.

- 900
- 1000
- 2200
- 2000

Q20. By what percent the number of positive tested people grew in Spain in February from January?

- 335.13%
- 235.13%
- 353.13%
- 253.13%

Section 2 - Verbal Ability

Section Summary

- No. of Questions: 10

- Duration: 10 min

Additional Instructions:
None

Q1. **Find the closest word which is similar to the given word :**

Recalcitrant

- Careless
- Dive
- Stubborn
- Casual

Q2. **Select the Synonym of the word**

WARY

- Sad
- Vigilant
- Distorted
- Tired

Q3. **Substitute the given sentences with one word.**

The **sudden damage and suffering** caused due to the Vizag Gas leak still have its effect on the mind, body and soul of local people.

- Antistrophe
- Apostrophe
- Catastrophe
- Anatomy

Q4. **In the following question, statements 1 and 6 are respectively the first and the last sentences of a paragraph and statements A, B, C and D come in between them. Rearrange A, B, C and D in such a way that they make a coherent paragraph together with statements 1 and 6. Select the correct order from the given choices and mark its number as your answer.**

1. Experts think that our planning has failed miserably.
A. Thus we have not succeeded in establishing a socialistic pattern of society.
B. A good percentage of our population is still living below the poverty line.
C. Thirty years of planning has failed to narrow the gap between the rich and the poor.

D. The rich have grown richer and the poor poorer.
6. Still we hope for a better planning and a better way of life.

- CBDA
- CBAD
- BCDA
- CDBA

Q5. **The sentences given in each question, when arranged in the proper sequence, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the given choices to construct a meaningful paragraph.**

(A) Yet few of the 500000 residents of the former Spanish colony are feeling flush with petrodollars; most have no plumbing or electricity.
(B) Although the coup failed, much remains at stake in the shaky oil state, where offshore drilling began in the early 1990s.
(C) "None of the oil revenues have gone into developing the country", says Sarah Wykes, a researcher for the London corporate watchdog group Global Witness.
(D) Production of about 250000 barrels per day is projected to nearly double in the next few years.

- CADB
- DBAC
- BDAC
- BDCA

Q6. **Choose the best possible answer for the following question:**

It's no coincidence that good guy/bad guy movies, comic books, and games have large, impassioned, and volatile _____.

- enthusiastic
- group
- fandoms
- zealous

Q7. **Fill in the blank with suitable options given below :**

Due to a phenomenon called subsidence, the metropolis's landscape is _____ and parts of the city are now dropping a foot and a half each year.

- compressed

compacting

widening

expanding

None of the mentioned options

Common Content:

Read the passage and answer the questions that follow.

We should make a distinction between foundations and principles, but not assume that they are wholly separate. The principles of good governance are derived from the foundations of humans which, in order to be truly universal, must themselves be established in the sacredness of life, and not just in some intellectual notion of what human order is. Principles that are derived wholly from mental constructs are, like them, shifting and like them, as easily abandoned. When they are replaced by other principles, the latter are constructs no less, and the cycle is repeated. It is true that each began with a deep concern for human progress, but it is also true that the thing was not long before put into jeopardy. That is because, perhaps, the successive ideas of human progress were so fragmented and limited to one's wrongly understood self-interest that they never became truly universal. Whatever else the history of government based on the principles of utilitarianism, liberalism and socialism shows, it shows that most of all. For it brought up again and again and always with reference to the actual living conditions of the people, the question of whose progress?

Dharma demonstrates how human rights, even before they are a matter of principle, are a matter of foundation - of a sane society. It will not be circular reasoning at all if it is said that in actual fact, they are a matter of Dharma. The literal meaning of the Sanskrit word 'dharma' is the foundation. The foundations or Dharma, of all relationships, personal and social, lie in the recognition that life is sacred, that all its expressions are inter-related and bound in visible unity, and that no violence shall be done to a person by another except with the due process of law, which at all times must be free from anger and hatred, and that the political and legal governance of human relationships shall have as their chief aim the enhancement of the individual and the progress of the people. It is from these that the principles of law and government must be derived, and it is in them that they shall have their ultimate validation.

This is the Dharmic law. Its concern with the dignity and worth of the individual and with social progress was given institutional forms of jurisprudence. That was done in the Mahabharata and some centuries later in the Arthasastra of Kautilya. Both are of great relevance to our times. But here let us have a brief look at the latter of the two, assigned to 321 B.C. in the reign of Chandragupta Maurya, and the discovery, of which, in 1905 by R.Shamasastry, was an event of very great excitement. Ahimsa, which I prefer to translate as "non-violence," the foundation of social existence, leads to the first principle of the law of punishment or danda-niti. It is stated at the very beginning of the Arthasastra."Whoever imposes severe punishment becomes repulsive to the people, while he who awards mild punishment becomes contemptible. But whoever imposes punishment as being deserved becomes respectable. For punishment, when awarded with due consideration, makes the people devoted to dharma and to works productive of wealth and joy; while punishment, when ill-awarded under the influence of greed and anger owing to ignorance, excites fury even among the heretics and ascetics dwelling in forests, not to speak of householders."

This principle then became the basis of the law against illegal confinement of any person, man or woman, and of the law against the unauthorised criminal trial of anybody. While in the custody of the police or of the jail superintendent, no accused or convict shall be insulted and humiliated by offensive language, nor beaten or tortured. Moreover, the principle of "punishment as deserved" must at all times be coupled with the principle of mercy. In such circumstances, for example, a diseased person anyone suffering from hunger, thirst, or fatigue from journey, a villager from country parts, anyone that has suffered much from punishment, and a moneyless pauper shall be shown mercy."In brief, the Arthasastra principle is that even an accused and a convict is entitled to his or her dignity. Whoever official will violate these principles, shall himself be liable to punishment.

In the case of a woman, her ill-treatment or rape by an officer, in whose custody she was placed either in the lock-up or in the prison was regarded as a serious offence and the officer was punished, taking into account the relative degree of the offence, either by fine or even by death. The Arthasastra further prescribed that if a woman was pregnant, or had just delivered a child, the sentence on her was to be kept in suspension until she was in a fit condition to be subjected to it.

The evident truth that woman is the spring of life, and her personality was not to be violated if life were to be regarded as sacred, led to the principle - "No man shall have sexual-intercourse with any woman against her will." Kautilya repeats this three times in a single chapter, and it clearly applied to the man-woman relationship in marriage as well. Whatever might be the position of the sastras of the medieval times on the question of divorce, the Arthasastra is perfectly clear that both man and woman have equal status in that."A woman, hating her husband, cannot dissolve her marriage with him against his will. Nor can a man dissolve his marriage with his wife against her will. But from mutual enmity, divorce may be obtained."

But the greatest significance is the Arthasastra's admonition to the king, now the state - "In the happiness of his subject lies his happiness, in their welfare, his welfare, he shall not consider as good to what pleases himself, but to what please his subjects. It was to this supreme maxim that all his public acts were subjected. And because dharma as justice is the foundation of civil society, then it led to

the principle that "the king shall with adequate punishments test first the conduct of government servants, and then only will examine the conduct of his people in towns and villages." For nothing is more destructive of social order than the evidence that the officers of the state are above the law.

For that reason, Kautilya places the conduct of judges under a discipline of the highest kind. To protect the people from unjust punishments, he prescribes the following. "When a judge threatens, browbeats, sends out or unjustly silences any one of the disputants in his court, if he does not ask what ought to be asked, or asks what ought not to be asked, leaves out what he himself has asked, makes unnecessary delay in, discharging his duties, compels the parties to leave the court by tiring them with delay, evades or causes to evade statements that lead to the settlement of a case, he shall be punished with the highest amercement. If he repeats the offence, he shall both be punished with double the above fine-and dismissed."

Q8. In 'Arthasastra' greatest prominence is given to

- the maxim that his children's welfare is the king's welfare.
- dharma as the foundation of social organization.
- the principle that people's happiness is the Kings's happiness.
- the waning or reproof by the king.

Q9. 'Dharma' shows that human rights, before becoming principles, make the foundation of a sensible society. Following is/are the intricacies of this phenomenon.

- the meaning of Dharma is foundation.
- human rights are an issue of Dharma.
- the foundations of all relationships are in knowing the sanctity of life.
- all its relationships are interrelated.
- All of these

Q10. Punishment awarded with one due care earns people's loyality to 'Dharma' and to creativeworks, when ill-awarded, earns

- contempt
- respect
- wrath
- criticsm

Section 3 - Advanced Coding

Section Summary

- No. of Questions: 2
- Duration: 50 min

Additional Instructions:
None

Q1. **Problem statement**

Given an array of integers. Find the Inversion Count in the array.
For an array, inversion count indicates how far (or close) the array is from being sorted. If the array is already sorted then the inversion count is 0. If an array is sorted in the reverse order then the inversion count is the maximum.
Formally, two elements a[i] and a[j] form an inversion if a[i] > a[j] and i < j.

Example
Input
5
2 4 1 3 5
Output
3
Explanation
Sequence 2, 4, 1, 3, and 5 has three inversions (2, 1), (4, 1), and (4, 3).

Input Format

The first line contains a single integer n, which represents the size of the array.
The second line contains n space-separated integers, which represent the elements of the array.

Output Format

The output consists of a single integer, which represents the number of inversions in the given array.

Refer to the sample input and output for formatting specifications.

Constraints

$1 \le N \le 5*10^5$
$1 \le arr[i] \le 10^{18}$

Sample Input	Sample Output
5 2 4 1 3 5	3

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Problem Statement**

The problem is to print all the possible paths from the top left to the bottom right of an m*n matrix with the constraints that from each cell you can either move only to the right or down. From each cell first, print all paths by going down and then print all paths by going right. Do this recursively for each cell encountered.

Example

Input:
2 3
1 2 3
4 5 6

Output:
1 4 5 6
1 2 5 6
1 2 3 6

Explanation:
Moving to the cells in this manner will give the right way to reach the destination from the starting point.

Input Format

The first line of input consists of two integers **m** and **n**, where **m** represents the number of rows in the matrix and **n** represents the number of columns in the matrix.
The next **m** lines consist of **n** integers each, representing the elements of the matrix.

Output Format

The output consists of all possible traversals of the matrix in the order of their occurrence separated by space.

Refer to the sample output for formatting specifications.

Constraints

In this scenario, the given test cases will fall under the following constraints:
1 ≤ m, n ≤ 5

Sample Input

```
2 2
1 2
3 4
```

Sample Output

```
1 3 4
1 2 4
```

Sample Input

```
2 3
1 2 3
4 5 6
```

Sample Output

```
1 4 5 6
1 2 5 6
1 2 3 6
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Answer Key & Solution

Section 1 - Advanced Quantitative Ability

Q1

70

Solution

The marks scored by various students are :

Hari = 75, Vikram = 65, Suresh = 65 + 45 = 110, Vinesh = 110-25 = 85, Ganesh = 85+34 = 119

The maximum marks = 119+50 = 169

Required percentage = (119/169) * 100 = 70% (approximately)

Q2

120%

Solution

Let the third no be 100

First number= 126

Second number= 105

126 is x% of 105

x= (126/105)*100= 120%

Q3

8.33%

Solution

Assume that cost price of book for Tarun = Rs. CTarun marks up the price of the book by 50%.∴The marked price = C + 50% of C =

1.5CTarun allows a discount of 20% and sells it to Diwana.The amount Tarun receives after providing discount = marked price – 20% of marked price= 1.5C – 20% of 1.5C= 1.2C

∴ Diwana purchased it for Rs. 1.2C

Diwana sells it for Rs. 20 more than what she purchased for.The selling price of book for Tarun = Rs. (1.2C + 20)

This S.P is 30% more than the original C.P of the toy car.⇒ 1.2C + 20 = C + 30% of C⇒ 0.1C = 20⇒ C = Rs. 200

Now, Diwana's profit % is:

% Profit =[(1.2C + 20) - 1.2C] / 1.2C = 0.0833 * 100 = 8.33

Q4

42.22%

Solution

Let CP be Rs. 3600 and MP be 100x

So, SP = 60x and Profit = Rs. 1200

So, SP = 3600 + 1200 = 60x

x = 80

So, discount = 40x = 40 * 80 = 3200

New discount = 3200 * 0.9 = Rs. 2880

So, New SP = 8000 - 2880 = Rs.5120

Hence, Profit % = (5120 - 3600)/3600 * 100% = 42.22%

Q5

308 cm^3

Solution

Volume of frustum of a cone = π/3 h (R^2 +r^2 + R x r)

Given, frustrum of a right circular cone has the radii of base 5 cm, of the top 3 cm and a height of 6 cm.

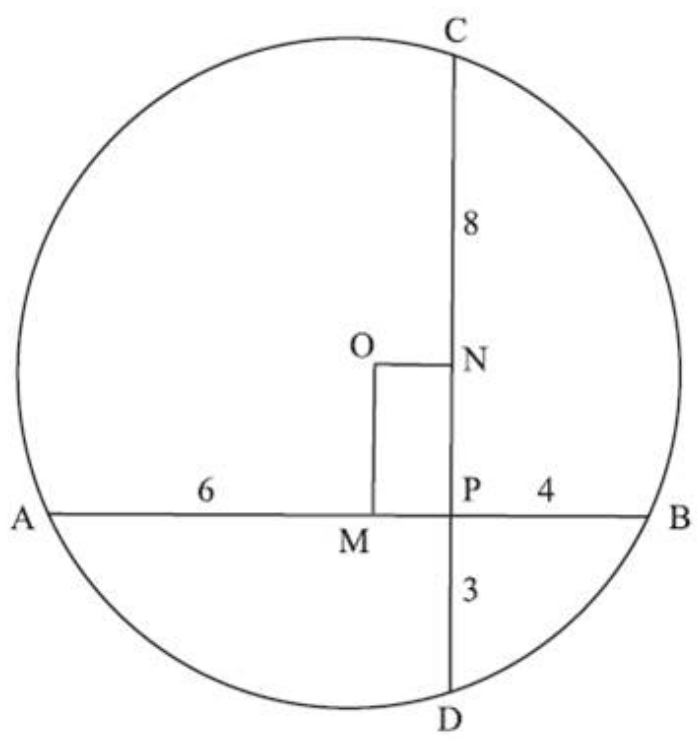
volume of frustrum = 22/7 x 1/3 x 6 x (5^2 + 3^2 + 5 x 3)

volume of the frustrum = 308 cm^3

Q6

125π/4 sq cms

Solution



As AB and CD are two chords that intersect at O, AP * PB = CP * PD

6 * 4 = CP * 3

CP = 8

From center O draw OM ⊥r AB and ON ⊥r CD.

From the center a line ⊥r to a chord bisects the chord.

So, we have AM = MB = 5 cm

MP = 1 cm, ON = 1 cm, CD = 11 cm, CN = 5.5 cm

$ON^2 + CN^2 = OC^2$

$1^2 + 5.5^2 = r^2$

$1 + 30.25 = r^2$

Area = πr^2

$\pi * 31.25$

$31.25\pi = 125\pi/4$ sq cms

Hence, the answer is $125\pi/4$ sq cms.

Q7

5759.52

Solution

Due to a fall in the interest rate of 0.5% the interest is reduced by 104 rupees.

0.05% of the Principal = 104

So P = 20800.

C.I for 2yrs at 13% intrest will be

C.I = $P(1 + r/100)^n - P$

Sub the values

C.I = 5759.52

Q8

Rs.4000

Solution

SI = Px20x1.5/100

CI = $P[(1+20/200)^3 - 1]$

$P[(1+20/200)^3 - 1]$ - Px20x1.5/100 = 124

P = 4000

Q9

Rs. 2500

Solution

Clark got 3000 so Chintu got 1500 .(The ratio becomes 2:1)

Now as Clark invested 10000 for 6 months and chintu invested x for the whole year so the ratio becomes:

(10000*6)/(x*12) = 2:1

Solving for x we get ⇒ 2500

Q10

2

Solution

Let A’ s present age = x years

Present age of wife = y years

Based on given data,

x = 5(x – 5)/4

∴ 4x = 5x – 25

⇒ x = 25

Also,

y + 4 = 3 (y – 5)/2 (given)

⇒ 2y + 8 = 3y – 15

∴ y = 23

Thus, difference in age is 2 years.

Q11

P = Rs. 440; Q = Rs. 385; R = Rs. 280

Solution

P= 7 days

Q= 8 days

R= 11 days

LCM= Total work= 7*8*11= 616 units

Efficiency of P= 88 units/day

Efficiency of Q= 77 units/day

Efficiency of R= 56 units/day

P share= (88/221)*Rs.1105= 440

Q share=(77/221)*Rs.1105= 385

R share = (56/221)*Rs.1105= 280

Q12

Rs.2500

Solution

A - 24 days = 5 units/day

B - 30 days = 4 units/day

Total work - 120 units

A and B working together for 10 days and completes 90 units

Remaining work = 30 units is completed by A

B completes 40 units

B'share = (40/120) x 7500 = 2500

Q13

144 kmph

Solution

R1 = (120 +120) / 16 = 240 / 16 = 15 m/s

R2 = (120 + 120) / 48/13 = 240 / 48/13 = 65 m/s

Speed of the faster train = 15 +65 / 2 = 80 / 2 = 40 m/s

= 40 x 18/5

= 144 kmph

Q14

13/30

Solution

A, B and C are 3 batsmen.

P(A) is probability of A hitting the target.

P(A') is probability of A not hitting the target.

Similarly for other marksmen.

P(A) = 1/3

P(A') = 1 - P(A)

= 2/3

Similarly for other batsmen, we have

P(B) = 1/4 ; P'(B) = 3/4

P(C) = 1/5 ; P'(C) = 4/5

Probability that one and only one of them hit the target is

P(A∩B'∩C') + P(A'∩B∩C') + P(A'∩B'∩C)

= (1/3 x 3/4 x 4/5) + (2/3 x 1/4 x 4/5) + (2/3 x 3/4 x 1/5)

= 13/30

Q15

7/10

Solution

The required probability will be given by

First is a place in Asia and Second is a place in Europe OR

First is a place in Europe and Second is a place in Asia OR

First is a Asia and Second is also a place in Asia

i.e. (7/16) x (9/15) + (9/16) x (7/15) + (7/16) x (6/15)

= 7/10

Alternatively, we can define the non-event as: There are two Asian places and no European places. Then, the probability of the nonevent is

(9/16) x (8/15) = 3/10

Hence, ***P(E) =*** (1– 3/10) = 7/10

Q16

60840

Solution

A sample plate might be, say

A T 8 3

There are 26 choices for a letter to go where the A is.

(Any of the 26 letters.)

There are then 26 choices for a letter to go where the T is.

(Any of the 26 letters.)

There are then 9 choices for a digit to go where the first 8 is.

(Any of the other 9 digits besides 0.)

There are then 10 choices for a letter to go where the 3 is.

(Any of the 10 digits, including 0.)

So we multiply all these together:

Hence 26*26*9*10 = 60840 different plates can be formed

Q17

209

Solution

We may have 1 student from age 12 group and 3 students from age 13 group (or) 2 students from age 12 and 2 students from age 13 (or) 3 students from age 12 and 1 student from age 13 (or) 4 students from age 12

required number of ways

$= ({}^6C_1 * {}^4C_3) + ({}^6C_2 * {}^4C_2) + ({}^6C_3 * {}^4C_1) + {}^6C_4$

$= ({}^6C_1 * {}^4C_1) + ({}^6C_2 * {}^4C_2) + ({}^6C_3 * {}^4C_1) + {}^6C_2$

$= 24 + 90 + 80 + 15 = 209$

Q18. 67.74

Solution

Number of people tested positive in March = 2100.

Number of days in March = 31

Average = 2100 / 31 = 67.74

Q19. 2200

Solution

Number of cases in India in January = 100

Number of cases in Japan in January = 2 × 100 = 200

Number of cases in February in India = 600

Number of cases in Japan in February = 600 + 50% of 600 = 900

Total cases in Japan till Feb end = 200 + 900 = 1100

Number of cases in March = 2 × 1100 = 2200

Q20. 235.13%

lution

umber of people found positive in January = 740

umber of people found positive in February = 2480

ercentage growth = 2480 – 740 / 740 x 100 = 235.13%

Section 2 - Verbal Ability

Q1

Stubborn

Solution

"Stubborn" is a synonym for "recalcitrant." Both words describe someone who is resistant to authority, control, or guidance and is unwilling to comply or cooperate.

Q2

Vigilant

Solution

To be wary means to be watchful or on guard; vigilant.

Q3

Catastrophe

Solution

The term "catastrophe" refers to a sudden and widespread disaster causing significant damage and suffering. It encapsulates the idea of a calamity that has profound and lasting effects on individuals.

Q4

CBDA

Solution

Statement 'C' gives an idea of what makes experts think that our planning has failed. Therefore, 'C' follows 1. 'B' follows 'C'. 'D' further explains 'C' and talks of 'rich arid poor'. The author gives his conclusion based on the above-stated evidence in statement 'A', which precedes statement 6.

Choice (1)

Q5

BDAC

Solution

Ans. (3). Statement B initiates the paragraph. D followed by A furthers the argument, by telling us that even though the projections show a near- doubling in the next two years, hardly any benefit of the offshore drilling activity in the state has been received by any of the

residents. A researcher comments that the oil revenues have not been directed towards progress of the country in the concluding statement C. Option (1): C and A though giving the same indication are separated by "Yet" which is contradictory. Option (2): D, which is a specific statement, precedes B, which is a general statement on the situation and should pave the way for the specifics, Option (4): Again, "Yet" of statement A contradicts the similar tone of C and A. Hence, the correct order is BDAC and the answer is (3).

Q6 fandoms

Solution

A noun would fill the blank which the adjectives 'lage, impassioned and volatile' modify so adjectvies 'enthusiastic' and 'zealous' can be eliminated.

Group is irrelevant as we talk about good guy/bad guy (an individual)

Hence, 'fandoms' is the answer which means being a fan

Q7 compacting

Solution

A verb would fill the blank because the auxiliary verb 'is' precedes the blank which is to be followed by the main verb.

Since the sentence is about subsidence (the gradual caving in or sinking of an area of land), 'widening' and 'expanding' are contextually unfit.

Also, the sentence is in present continuous tense as suggested by 'dropping', so compressed is eliminated.

Hence, the answer is compacting.

Q8. the principle that people's happiness is the Kings's happiness.

Solution

The tenth paragraph, in the opening line, states and upholds option (the principle that people's happiness is the Kings's happiness.) as the correct one. Other options are not correct in the given context;

Hence, option (3the principle that people's happiness is the Kings's happiness.) is the correct option.

Q9. All of these

lution

Options:

1. The meaning of Dharma is foundation. ✅ **True**
2. Human rights are an issue of Dharma. ✅ **True**
3. The foundations of all relationships are in knowing the sanctity of life. ✅ **True**
4. All its relationships are interrelated. ✅ **True**

All four statements are explicitly supported by the passage.

✅ **Correct answer: All of the above**

Q10. wrath

Solution

The sixth paragraph towards the end endorses the option "wrath".

It is the option sought. Wrath means extreme anger.

Section 3 - Advanced Coding

Q1

Hidden Test Case

Input

```
5
2 3 4 5 6
```

Output

```
0
```

Weightage - 10

Input

```
3
10 10 10
```

Output

```
0
```

Weightage - 25

Input

```
10
1 5 4 7 85 6 3 2 1 5 4
```

Output

```
24
```

Weightage - 25

Input

```
15
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

Output

```
0
```

Weightage - 40

Sample Input

```
5
2 4 1 3 5
```

Sample Output

```
3
```

Solution

```
import java.util.*;
import java.io.*;
import java.lang.*;

public class Main
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner(System.in);
            long n = sc.nextLong();
            long arr[] = new long[(int)n];

            for(long i = 0; i < n; i++)
             arr[(int)i] = sc.nextLong();

            System.out.println(new Sort().inversionCount(arr, n));

    }
}
//User function Template for Java

class Sort
{
    static long count=0;
    static long inversionCount(long a[], long N){
     int n=(int)N;
     long ct=count;
     mergeSort(a,0,n-1);
     return count-ct;
    }

    public static long[] mergeSort(long[] a, int low, int high){
    if(low==high){
        long[] single=new long[1];
        single[0]=a[low];
        return single;
    }
    int mid=(low+high)/2;
    long[] left=mergeSort(a,low,mid);
    long[] right=mergeSort(a,mid+1,high);
    long[] merged=merge2SortedArrays(left,right);
    return merged;
    }

    public static long[] merge2SortedArrays(long[] left, long[] right){
    int i=0,j=0,k=0;
    long[] merging=new long[left.length+right.length];
    while(i<left.length && j<right.length){
        if(left[i]<=right[j]){
            merging[k]=left[i];
            i++;
            k++;
        }
```

```
            else{
                count+=left.length-i;
                merging[k]=right[j];
                j++;
                k++;
            }
        }
        while(i<left.length){
            merging[k]=left[i];
                i++;
                k++;
        }
        while(j<right.length){
            merging[k]=right[j];
                j++;
                k++;
        }
        return merging;
    }
}
```

Q2

Hidden Test Case

Input

```
3 2
1 3
5 7
9 11
```

Output

```
1 5 9 11
1 5 7 11
1 3 7 11
```

Weightage - 10

Input

```
3 3
1 2 3
4 5 6
7 8 9
```

Output

```
1 4 7 8 9
1 4 5 8 9
1 4 5 6 9
1 2 5 8 9
```

Weightage - 15

Input

```
3 2
1 2
4 5
7 8
```

Output

```
1 4 7 8
1 4 5 8
1 2 5 8
```

Weightage - 10

Input

```
5 5
1 2 3 10 11
4 5 6 12 13
7 8 9 14 15
```

Output

```
1 4 7 16 19 23 24 25 22
1 4 7 16 21 23 24 25 22
1 4 7 16 21 17 24 25 22
1 4 7 16 21 17 20 25 22
```

Weightage - 15

Input

```
5 5
1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
```

Output

```
1 6 11 21 16 17 18 19 20
1 6 11 21 22 17 18 19 20
1 6 11 21 22 23 18 19 20
1 6 11 21 22 23 24 19 20
```

Weightage - 25

Input

```
4 5
1 2 3 4 5
9 8 7 6 5
4 8 2 9 5
```

Output

```
1 9 4 0 2 4 7 1
1 9 4 8 2 4 7 1
1 9 4 8 2 4 7 1
1 9 4 8 2 9 7 1
```

Weightage - 25

Sample Input

```
2 2
1 2
3 4
```

Sample Output

```
1 3 4
1 2 4
```

Sample Input

```
2 3
1 2 3
4 5 6
```

Sample Output

```
1 4 5 6
1 2 5 6
1 2 3 6
```

Solution

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
class MatrixTraversal {

    private static void printMatrix(int[][] mat, int m, int n, int i, int j, List<Integer> list) {
        if (i > m || j > n)
            return;

        list.add(mat[i][j]);

        if (i == m && j == n) {
            for (int k = 0; k < list.size(); ++k) {
                if (k > 0) {
                    System.out.print(" ");
                }
                System.out.print(list.get(k));
            }
            System.out.println();
        }

        printMatrix(mat, m, n, i + 1, j, list);
        printMatrix(mat, m, n, i, j + 1, list);

        list.remove(list.size() - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int m = sc.nextInt();
        int n = sc.nextInt();

        int[][] mat = new int[m][n];

        for (int i = 0; i < m; i++) {
            for (int j = 0; j < n; j++) {
```

```
                mat[i][j] = sc.nextInt();
            }
        }

        List<Integer> list = new ArrayList<>();
        printMatrix(mat, m - 1, n - 1, 0, 0, list);
    }
}
```

```
def printMatrix(mat, m, n, i, j, lst):
    if i > m or j > n:
        return

    lst.append(mat[i][j])

    if i == m and j == n:
        for k in range(len(lst)):
            if k > 0:
                print(" ", end="")
            print(lst[k], end="")
        print()

    printMatrix(mat, m, n, i + 1, j, lst)
    printMatrix(mat, m, n, i, j + 1, lst)

    lst.pop()

if __name__ == "__main__":
    m, n = map(int, input().split())

    mat = []
    for i in range(m):
        row = list(map(int, input().split()))
        mat.append(row)

    lst = []
    printMatrix(mat, m - 1, n - 1, 0, 0, lst)
```

```
#include <stdio.h>
#include <stdlib.h>

void printMatrix(int **mat, int m, int n, int i, int j, int *list, int *size) {
    if (i > m || j > n) {
        return;
    }

    // Add current element to the list
    list[(*size)++] = mat[i][j];

    // If we reach the bottom-right corner, print the path
    if (i == m && j == n) {
        for (int k = 0; k < *size; ++k) {
            if (k > 0) {
                printf(" ");
            }
            printf("%d", list[k]);
        }
        printf("\n");
    }

    // Traverse the matrix by going down and then right
    printMatrix(mat, m, n, i + 1, j, list, size);
```

```
    printMatrix(mat, m, n, i, j + 1, list, size);

    // Backtrack: Remove the last element from the list
    (*size)--;
}

int main() {
    int m, n;
    scanf("%d %d", &m, &n);

    // Create the matrix dynamically
    int **mat = (int **)malloc(m * sizeof(int *));
    for (int i = 0; i < m; i++) {
        mat[i] = (int *)malloc(n * sizeof(int));
    }

    // Read the matrix elements
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &mat[i][j]);
        }
    }

    // Prepare an array to store the current path
    int *list = (int *)malloc((m + n) * sizeof(int)); // Maximum size of the path list
    int size = 0; // Keeps track of the size of the current path

    // Start the traversal from the top-left corner (0, 0)
    printMatrix(mat, m - 1, n - 1, 0, 0, list, &size);

    // Free the dynamically allocated memory
    for (int i = 0; i < m; i++) {
        free(mat[i]);
    }
    free(mat);
    free(list);

    return 0;
}
```

```
#include <iostream>
#include <vector>
using namespace std;

void printMatrix(int **mat, int m, int n, int i, int j, vector<int>& list) {
    if (i > m || j > n)
        return;
    list.push_back(mat[i][j]);
    if (i == m && j == n) {
        for (int k = 0; k < list.size(); ++k) {
            if (k > 0) {
                cout << " ";
            }
            cout << list[k];
        }
        cout << endl;
    }
    printMatrix(mat, m, n, i + 1, j, list);
    printMatrix(mat, m, n, i, j + 1, list);
    list.pop_back();
}

int main() {
    int m, n;
```

```
    cin >> m >> n;
    int **mat = new int*[m];
    for (int i = 0; i < m; ++i) {
        mat[i] = new int[n];
        for (int j = 0; j < n; ++j) {
            cin >> mat[i][j];
        }
    }
    vector<int> list;
    printMatrix(mat, m - 1, n - 1, 0, 0, list);

    for (int i = 0; i < m; ++i) {
        delete[] mat[i];
    }
    delete[] mat;
    return 0;
}
```